EXPERIMENT 09

Rule-Based POS Tagging Using Regular Expressions

Aim

To implement a rule-based part-of-speech tagger using regular-expression patterns.

Procedure

1. Import RegexpTagger from NLTK.
2. Define suffix-based tagging patterns.
3. Tokenize a sample sentence.
4. Apply the regular-expression tagger.
5. Print the tagged tokens.

Program

```
# Install first: pip install nltk
from nltk import RegexpTagger, word_tokenize

patterns = [
    (r".*ing$", "VBG"),
    (r".*ed$", "VBD"),
    (r".*ly$", "RB"),
    (r".*s$", "NNS"),
    (r".*", "NN")
]
tagger = RegexpTagger(patterns)
sentence = "Birds are flying quickly".split()
print(tagger.tag(sentence))
```

Output

```
OUTPUT — EXPERIMENT 09
[('Birds', 'NNS'), ('are', 'NNS'), ('flying', 'VBG'), ('quickly', 'RB')]
```

Result

Thus, regular-expression rules were used to assign approximate POS tags to words.